

Problem Background

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. SIH problem context

The SIH problem is Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines, under Ministry of Steel / NMDC. The ProofRoad document states that Bailadila monsoon conditions can reduce visibility to roughly 3–5 m and that loaded dumpers may need much more distance to stop. (ProofRoad pp. 1, 5).

The externally reproduced SIH problem statement describes the operational effect as slower or halted dumper movement, increased haul cycle times, reduced fleet productivity, lower ore evacuation and production losses, together with increased collision and road-accident risk. A community-maintained SIH archive and other secondary reproductions carry the same problem wording; treat the uploaded proposal as the primary project framing.

External context: the Met Office explains that modern visibility observations can be instrument-based and defines meteorological optical range in terms of attenuation of a light beam. This supports the idea of measuring visibility rather than relying only on driver perception.

2. Operational environment

Condition / factor	Effect on operation	ProofRoad response
Dense fog / 3–5 m visibility	Driver visual horizon collapses	Radar-first sensing; unknown road is never treated as clear
Wet downhill haul road	Braking performance worsens	Use pessimistic tested braking value and slope/wet-road penalties
Blind curve	Future road occupancy is uncertain	Fail-closed gate logic; uncertain means wait
Network loss	Peer information becomes stale	Own radar and safety MCU continue independently
Sensor failure	Perception coverage decreases	Failure never increases proven-clear distance

ProofRoad pp. 3–5, 9–10.

3. Why the problem is technically difficult

The core difficulty is not simply detecting fog. The system must decide how much road can be treated as proven clear, compare that distance with the stopping distance required by the current truck state, and remain conservative when evidence becomes unavailable.

Dense fog also creates asymmetric sensor performance. ProofRoad assigns radar the authority to prove the road clear, while RGB, thermal and NIR can only shorten the proven distance. This is a deliberately conservative fusion architecture (p. 4).

The problem also has a fleet dimension. A truck may know about a hazard before its own sensors can see it if another truck reports it. ProofRoad therefore uses small position-tagged messages and a Phantom Obstacle marker (pp. 3, 6–7).

4. Research questions

How far can 60/77 GHz radar reliably detect relevant mine-road obstacles under fog, rain and dust?

How should safe speed be computed from proven-clear distance, reaction time, braking capability, slope and road condition?

How should stale, missing, forged or contradictory V2V/V2I messages affect the safety state?

How can the system demonstrate productivity improvement without making unsupported safety claims?

What hardware, calibration and functional-safety evidence would be required before field deployment?

5. Key measurable variables

Visibility (m), radar detection range (m), obstacle range error, detection latency, false-alarm rate, missed-detection rate, braking estimate (m/s^2), reaction-time assumption (s), buffer distance (m), safe-speed advice (km/h), message age, packet loss, gate state, near-miss count, warning rate and haul-cycle time.